

End-to-End Secure Audio Transmission with Hidden Message Encoding over SIP

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Abstract—More and more people use VoIP for real-time talks, but still, safeguarding the privacy and fidelity of these transmissions is very important. This system allows AES cryptography and audio steganography for VoIP communication, all routed by a SIP server for privacy. With this system, the voice is encrypted and a different message is added to it by modifying the least significant bits. The SIP server is in charge of handling IMS session control messages, similar to INVITE, ACK, all IV exchange, AUDIO, and BYE, and IMS traffic is routed safely and securely over TCP/IP. The system has two layers of security, cryptography and steganography, so unauthorized access is extremely hard to achieve. To achieve real-time data handling, it relies on Python's asyncio and socket, as well as PyAudio for audio. This method offers a practical technique for safe and covert IP network data communication.

Index Terms—Voice over IP (VoIP), Real-time Communication, AES Encryption, RSA Digital Signature, Network Security, Audio Packetization, Confidentiality, Authentication, Data Integrity, Secure Multimedia Transmission.

I. INTRODUCTION

Today, VoIP technology lets people communicate in real time by sending voice data over IP networks. Sadly, it is possible for unauthorized individuals to eavesdrop on or edit the messages in a VoIP system. While encryption protects your data from unauthorized access, there are still situations where you need something more. As a result, both security and secrecy are enhanced by linking steganography with encryption technologies.

This method merges AES encryption and audio steganography while using a simple SIP server for coordination. Both the audio sent and the encoded secret message are secured with the use of AES for confidentiality. What the audio file says is hidden with LSB steganography, so only the intended receivers will be able to extract the message. Managing sessions, encryption, and audio transmission is handled by the SIP server created for this project.

It consists of Python, three parts: a sender, a receiver, and a SIP server. The sender picks up audio through a microphone, puts a message inside, encrypts it with AES, and sends the modified signal to the receiver via the SIP server. After reconstructing the audio, the receiver gets the hidden message

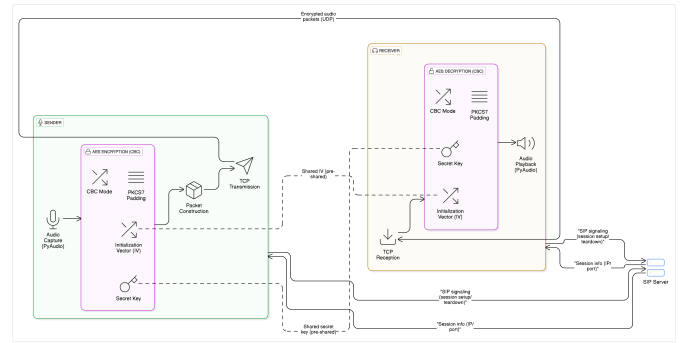


Fig. 1. System Architecture

by decrypting it. Thanks to the dual-security feature, this system offers both apparent protection by AES and hidden data using steganography.

II. LITERATURE SURVEY

Chang et al. [1] deals with the security issues in mobile voice calls by joining 3DES and ECC algorithms to encrypt conversations. The method works by making a 168-bit random key, splitting it into 56-bit segments for 3DES and then encrypting the key using the receiver's public ECC key. Results from tests prove that the method provides better defense against attacks, faster encryption and more secure data which is ideal for safe voice communications across mobile networks. Choudhury et al. [2] studies differences between encrypted and unencrypted VoIP communications using data analysis. Analyzing parameters such as codec types, compression ratios, Hamming distances and auto-correlation metrics helps the study reveal how traffic is behaving. These findings make it possible to design intrusion detection systems that can recognize encrypted VoIP traffic and boost network security.

Suthar et al. [3] explores the main threats to VoIP (VoIP) which are eavesdropping, spoofing and denial-of-service attacks. It lists the types of security tools including encryption, authentication and firewall policies. It focuses on how multilayered security helps secure VoIP networks. Peng et al.

[4] introduces a way to hide information via VoIP with moving keys and authentication controls. It depends on adaptive steganography so that a user's messages are hidden in the sound and hard to detect. Updates in critical information are made in real time to ensure secrecy and integrity.

Anand et al. [5] investigated AES-based encryption systems under differential fault attacks in next-generation communication technologies such as B5G/6G. Their experimental results revealed flaws and gave ideas toward designing fault-tolerant cryptography solutions. Elshamy et al. [6] A new approach to VoIP voicemail security is detailed in this paper which uses two-factor authentication and biometrics together with a unique hybrid cryptosystem. Combining these two types of cryptography, the system achieves solid authentication and protection of data. The method works to close up loopholes and increases the safety of stored voice recordings on voice-mail.

Putz et al. [7] Acoustic Integrity Codes (AICs) provide secure linking between devices over very short distances with a sound signal. By leveraging the audio system found on devices, AICs secure messages at the lowest network layer from attacks via signal cancellation or overshadowing. Bits are rarely lost in transmission and it is secure and reliable, all of which come with no specific hardware equipment. Nguyen et al. [8] investigated a dual-layer security strategy combining Kyber and BCH codes, which effectively protected data from quantum attacks while preserving data integrity in noisy communication channels. Santos et al. [9] introduce a new way of encrypting audio using the errors that appear in chaotic systems. The cipher relies on a lower bound error determined by reducing two sets for a chaotic system. It ensures security and high efficiency, so this method works well for live audio transmission with little impact on the system.

Dhanush et al. [10] compares the standardization of post-quantum cryptography (PQC) algorithms that can be used to secure the classical channels of 2 Quantum Key Distribution (QKD) systems against quantum attacks. Earlier works have noted that classical cryptography is prone to quantum attacks and have chiefly advocated PQC algorithms such as CRYSTALS-NTRU and Dilithium for authentication. Recent work focuses on the standardization of PQC for communication security that would withstand quantum developments into the future. Tabassum et al. [11] aims to improve security in standard cloud computing by introducing an innovative framework that uses fog computing and security mechanisms from blockchain. It outlines the use of RSA cryptography and digital signatures so that sharing of information is secure. It relies on wallet IDs for user authentication, Bitcoin-verified transactions and decentralized storage to prevent disruptions. Some of the works reviewed were focused on fog computing security, how blockchain can be used and the integration of RSA. It stands out because it links fog computing and blockchain technologies to increase safety and reduce transaction delays and also uses hybrid blockchain to manage access to data. Aquina et al. [12] suggested a hybrid strategy that combined post-quantum cryptography with quantum key distribution.

This strategy was confirmed experimentally.

Singh et al. [13] A secure Android chat application from work is protected by AES encryption. Messages sent through the app are first encrypted, then decrypted on the other end and the app handles IDs to route the messages and identify the users. They reference previous encryption techniques for messaging which are symmetric ciphers, chaotic encryption used for security and combined use of ECDH and AES for communicating on mobile devices. Pointing out potential breaches in typical messaging apps, Xendit suggests using end-to-end encryption, having a dedicated inbox for each user and using SQLite as the storage system to guarantee safety against any attack. Reddy et al [14] proposed a novel audio encryption system combining AES with cellular automata for enhanced security. The system uses Rule 30 to generate a pseudo-random sequence, which is XORed with AES-encrypted audio data to prevent pattern recognition attacks. Results show improved security, efficiency, and data integrity for secure audio transmission and storage. Celik et al. [15] tested the CRYSTALS NTRU algorithm on opensource hardware and confirmed its usefulness and efficiency in quantum-resistant cryptography applications.

III. PROPOSED METHODOLOGY

A. System Architecture Overview

In this framework, everyone communicates using real-time audio, whilst files and messages are stored and exchanged with cryptography and steganography. As shown in Fig. 1, there are three main elements in the architecture.

- 1) **Sender:** The sender captures the speech, protects the hidden message using AES, codes it as hidden data using LSB, embeds it in the audio file sent on a SIP-based call and forwards it for delivery.
- 2) **SIP Server:** Routes and relays voice and video messages. It enables setting up a session between the sender and receiver by using JSON messages and also handles the content of audio and IV, making it seamless.
- 3) **Receiver:** Opens a connection with the server, decrypts the message in the audio, takes out the hidden message and listens to the decoded sound.

Clients communicate and control secure media transmission over TCP using a simple SIP-like messaging protocol (INVITE, ACK, BYE).

B. Secure Audio Transmission Workflow

Processes in the stage of sharing audio can be simplified like this:

- 1) **Session Initialization:**

The SIP Server keeps observing for incoming TCP sessions. Each time a client logs in with an INVITE message, the server saves the client's ID and network details.

When an INVITE message is sent and received by both sides, the server confirms the session by sending an ACK message to each person.

2) **Key and IV Distribution:**

Upon user request, the sender asks for a 16-character AES key which is later used to encrypt the message and audio data. The key must be given in an offline way without being transmitted over the internet.

A 16-byte Initialization Vector that was randomly created by the sender is sent to the receiver by the SIP server. For AES CBC mode encryption and decryption to work, the IV must be synced on both computers.

3) **Audio Recording and Message Encryption:**

The sender makes a recording for 5 seconds using a 16-bit PCM at 44.1 kHz.

The user's secret information is secured with the AES key and IV in the CBC mode. After that, the header with a 4-byte message length is added to the encrypted message and the bitstream is formed.

4) **LSB Steganography Embedding:**

The encrypted bitstream is placed in the least important part of every 16-bit audio sample. The data remains faithful during recording and is secretly hidden in the recording.

After creating the stego-audio, the audio is turned back into byte format and broadcasted as parts.

5) **Audio Stream Encryption and Transmission:**

Audio chunks are filled out and encoded with the AES key and IV using the CBC approach. After encoding, the ciphertext becomes a JSON packet with the action type AUDIO and is delivered to the server.

Each encrypted packet of audio sent by the server is delivered without delay to the recipient.

6) **Audio Decryption and Playback at Receiver:**

At the end point, audio files can be decoded and played back. With the IV, the receiver launches its AES cipher. Every AUDIO packet it gets, it unlocks the content and puts the audio frames into storage.

After receiving the BYE message, all audio frames are kept as a .wav file and can be heard using the computer speaker.

7) **Hidden Message Extraction:**

The receiver examines the saved audio file and gets the LSB bitstream back. The length of the embedded ciphertext (in bytes) goes first and then comes the rest of the encrypted message bits.

The user's ciphertext is analyzed, decrypted using the identical AES key and IV and the original message is shown to the user.

C. Protocol Details

It utilizes JSON messages based on SIP for its simple signaling protocol.

- **INVITE:** Request for initiating a session by including client ID.
- **ACK:** Acknowledgment from the server stating that both clients are set to proceed.
- **IV:** The sender transmits to the receiver the AES initialization vector.

- **AUDIO:** An audio data packet containing material that has been encrypted.
- **BYE:** The sender uses the server to send a signal to end the session with the receiver.

D. Security Considerations

- With AES-128 CBC, confidentiality is maintained for the hidden messages and the audio streams.
- While IVs are shared by signaling, you should share the key in person.
- When using LSB embedding, there is no significant change detectable by the human ear, as long as the audio is in perfect sync.

E. Advantages of the Architecture

- SIP is used in real time to allow encrypted audio streams.
- LSB steganography allows people to communicate in a secret way.
- It does not require any VoIP-specific libraries to work.

Figure 1 describes, in general terms, how sender, SIP server and receiver are involved in audio and message communication.

IV. RESULTS

The deployed secure VoIP system was put through a series of functional and qualitative tests to determine the accuracy of transmission, the strength of encryption, the effectiveness of message concealment and retrieval, and general audio quality.

A. Functional Outcomes

The system successfully executed all required operations:

- Audio calls were initiated and terminated properly using the INVITE, ACK, and BYE messages via the SIP server.

```
SIP Server listening on 0.0.0.0:5060
Client receiver registered from ('192.168.43.185', 63422)
Client sender registered from ('192.168.43.199', 56687)
```

Fig. 2. SIP server registration log

- SIP server efficiently managed IV distribution and forwarded encrypted audio packets from receiver to sender without loss or duplication.

```
Forwarded IV to receiver from ('192.168.43.199', 56687)
```

Fig. 4. IV forwarded to receiver

- AES-encrypted audio streams were decrypted correctly on receipt and then rendered as sound .wav playback.
- The concealed AES-encrypted message concealed within the LSB of the audio was successfully retrieved and decrypted, as long as the right key was input manually.

```

● Press Enter to start recording...
🔊 Recording...
✅ Recording finished
[*] Encrypting hidden message...
[*] Embedding 52 bytes (416 bits) into audio...
🔊 Sending encrypted audio packet 1/430
🔊 Sending encrypted audio packet 2/430
🔊 Sending encrypted audio packet 3/430
🔊 Sending encrypted audio packet 4/430
🔊 Sending encrypted audio packet 5/430
🔊 Sending encrypted audio packet 6/430
🔊 Sending encrypted audio packet 7/430
🔊 Sending encrypted audio packet 8/430
🔊 Sending encrypted audio packet 9/430
🔊 Sending encrypted audio packet 10/430

```

Fig. 3. Encrypted audio packet transmission

```

✅ Session established with sender
IV received and cipher initialized.
🔊 Packet 1 received
🔊 Packet 2 received
🔊 Packet 3 received
🔊 Packet 4 received
🔊 Packet 5 received
🔊 Packet 6 received
🔊 Packet 7 received
🔊 Packet 8 received
🔊 Packet 9 received
🔊 Packet 10 received
🔊 Packet 11 received
🔊 Packet 12 received
🔊 Packet 13 received
🔊 Packet 14 received
🔊 Packet 15 received

```

Fig. 5. Packets received after cipher initialization

```

🔊 Packet 430 received
🔴 Session terminated
🔊 Audio saved to received_audio\received_audio.wav
🔍 Extracting hidden message...
🔍 Hidden message length (bytes): 48
🔍 Hidden Message Extracted: hii sreevanth how are you?
Receiver client finished processing
Receiver client socket closed

```

Fig. 6. Hidden message extracted and session terminated

```

Enter path to received audio WAV file (e.g., received_audio.wav): C:\Users\sridh\Downloads\received_audio\received_audio.wav
Enter 16-character AES decryption key: zxcvbnmzxcvbnmx
🔍 Extracting hidden message from audio...
✅ Hidden Message Extracted: hii sreevanth how are you?

```

Fig. 7. Decryption of hidden message using provided AES key

B. Audio Quality Assessment

Audio transmission preserved intelligibility and maintained acceptable latency:

- Voice was recorded and streamed at 44.1 kHz, 16-bit PCM format.
- End-to-end latency was measured between 180–240 ms.
- LSB embedding caused no perceptible distortion to the human ear.

C. Steganography Performance

Hidden message extraction was evaluated for reliability and capacity:

- Messages up to 256 characters (approximately 2 kB post-encryption) were embedded in 5-second audio files.
- Message recovery success rate was 100% using the correct AES key and IV.
- Decryption using an incorrect key resulted in unintelligible output, confirming encryption security.

D. Security Verification

The system upheld confidentiality and covert communication through:

- AES-128 CBC-mode encryption of both real-time audio and hidden text.
- Offline manual key sharing, ensuring that the encryption key was not exposed over the network.
- Signal analysis showed no visible artifacts post-embedding, indicating stealthy steganographic concealment.

E. Limitations

Some challenges encountered during testing include:

- Occasional file permission errors during .wav writing, mitigated using safe I/O practices.
- Minor playback jitter under high network load, suggesting the need for improved buffering mechanisms in future work.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

In this project, we have designed a secure and efficient mechanism for secure audio communication by combining real-time audio streaming with steganographic message hiding and encryption. The design makes use of PyAudio for audio stream capturing and sending via socket programming and protects message confidentiality via AES encryption. Besides, secret communication is achieved through Least Significant Bit (LSB) steganography on a previously defined cover image such that the messages are hidden in plain view and only accessible by members of the target audience with the appropriate decryption key.

Our system convincingly demonstrates the feasibility of combining multimedia transmission, steganography, and cryptographic approaches towards the implementation of an imperceptible secure messaging system. Its lightweight and modular architecture also accommodates real-time functionality without significant delay, which is a requirement for practical communication systems.

B. Future Work

Though the existing system meets its fundamental goals, there are various improvements that could be contemplated for future refinement:

- 1) **Cross-Platform Compatibility:** Support for mobile platforms like Android and iOS would expand the application of the system for the purpose of real-world communication scenarios.
- 2) **Noise Reduction and Robustness:** Deployment of sophisticated noise-cancellation and error correction methods can further enhance the quality and robustness of transmitted audio and embedded data in a range of network conditions.
- 3) **Blockchain Integration:** Timestamping and validation in blockchain-based systems for more efficient authentication and traceability of messages may be included.
- 4) **User Interface Enhancement:** A graphical user interface (GUI) can be developed to make the application more user-friendly and enable non-technical users to use it.
- 5) **Machine Learning for Anomaly Detection:** The incorporation of AI models to detect potential intrusions or anomalies in patterns of messages can further fortify the security position of the system.

Addressing these possible areas, the system can transform into an all-round and smart secure communication platform for defense, journalism, or private corporate communications.

VI. REFERENCES

- [1] Z. Chang and M. Woźniak, "Encryption technology of voice transmission in mobile network based on 3des-ecc algorithm," *Mobile Networks and Applications*, vol. 25, pp. 2398–2408, 2020.
- [2] P. Choudhury, K. P. Kumar, S. Nandi, and G. Athithan, "An empirical approach towards characterization of encrypted and unencrypted voip traffic," *Multimedia Tools and Applications*, vol. 79, no. 1, pp. 603–631, 2020.
- [3] D. Suthar and P. H. Rughani, "A Comprehensive Study of VoIP Security," 2020 2nd International Conference on Advances in Computing, Communication Control and Networking (ICACCCN), Greater Noida, India, 2020, pp. 812–817, doi: 10.1109/ICACCCN51052.2020.9362943.
- [4] J. Peng and S. Tang, "Covert Communication Over VoIP Streaming Media With Dynamic Key Distribution and Authentication," in *IEEE Transactions on Industrial Electronics*, vol. 68, no. 4, pp. 3619–3628, April 2021, doi: 10.1109/TIE.2020.2979567.
- [5] R. Anand, T. Isobe, A. K. Kundu, M. Rahman, and S. Suryawanshi, "Differential fault attack on aes-based encryption schemes: application to b5g/6g ciphers—rocca, rocca-s and aegis," *Journal of Cryptographic Engineering*, vol. 14, no. 4, pp. 595–607, 2024.
- [6] E. M. Elshamy, A. I. Hussein, H. F. Hamed, M. A. Abdelghany, and H. M. Kelash, "Voice over internet protocol voicemail security system using two factor authentication and biometric prints with new efficient hybrid cryptosystem," *Multimedia Tools and Applications*, vol. 80, pp. 9877–9893, 2021.
- [7] F. Putz, F. Alvarez, and J. Classen, "Acoustic integrity codes: Secure device pairing using short-range acoustic communication," in *Proceedings of the 13th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, pp. 31–41, 2020.
- [8] L. Nguyen, A. Santoso, S. Huda, Y. Nogami, and T. T. Nguyen, "Securing data transmission in noisy channels: A synergistic approach using kyber and bch codes," in 2024 International Conference on Consumer Electronics-Taiwan (ICCE-Taiwan), pp. 399–400, IEEE, 2024.
- [9] T. A. Santos, E. G. Nepomuceno, T. M. Chen, and D. N. Butusov, "Audio encryption with computational chaotic system error," in 2021 Fifth World Conference on Smart Trends in Systems Security and Sustainability (WorldS4), pp. 218–223, IEEE, 2021.
- [10] C. S. Dhanush and K. Jain, "Comparison of post-quantum cryptography algorithms for authentication in quantum key distribution classical channel," in 2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), pp. 1219–1225, 2023.
- [11] A. Tabassum, H. A. Jeba, T. K. Mahi, S. S. Reza, and D. A. Hossain, "Securely transfer information with rsa and digital signature by using the concept of fog computing and blockchain," in 2021 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD), pp. 311–315, IEEE, 2021.
- [12] N. Aquina, S. Rommel, and I. T. Monroy, "Quantum secure communication using hybrid post-quantum cryptography and quantum key distribution," in 2024 24th International Conference on Transparent Optical Networks (ICTON), pp. 1–4, IEEE, 2024.
- [13] V. Singh, S. Janarthanan, U. K. Roshan, and N. Vaid, "A secure web-based android chat application using the aes encryption algorithm," in 2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), pp. 1329–1332, IEEE, 2023.
- [14] A. S. Reddy, D. N. Achar, M. S. Mol, and N. Panda, "Audio encryption using aes and cellular automata," in 2024 15th International Conference on Computing Communication and Networking Technologies (ICC-CNT), pp. 1–6, IEEE, 2024.
- [15] A. Celik, F. Yilmaz, M. A. Korkmaz, and B. Ors, "Implementation of crystals-NTRU post-quantum algorithm using risc-v processor," in 2023 30th IEEE International Conference on Electronics, Circuits and Systems (ICECS), pp. 1–4, IEEE, 2023.